AirSky Airlines

An Imaginary Airline Company

DATA VISUALIZATION TECHNIQUES CRN-15360-202301

Group 8

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AirSky Airlines

Introduction:

AirSky is newly established airline in the American aviation industry, connecting people and destinations with a commitment to innovation, and unparalleled service. Established in 2023 in North America, AirSky has evolved into a cornerstone of air travel, operating a vast fleet of state-of-the-art aircraft.

Mission:

AirSky is dedicated in providing reliable and seamless travel to our customers. We are committed to minimize the flight delays and bring our service to as many customers as possible.

Functional area of interest:

We are committed to provide state-of-the-arc Airline Operations. We utilize advanced data visualization techniques to optimize our flight operations and minimize air traffic delay, at the same time maximizing our reach to as many passengers as possible fulfilling their transportation needs.

Topic of Case Study:

Maximize reach to passengers for on-time flight service.

Objectives and Scope of Case Study:

- To avoid flight delays by identifying the root cause of delays using state-of-the-arc data visualization technology.
- To understand potential passengers and provide flight services satisfying transportation demands.
- Develop marketing campaigns to attract customers during off-peak periods.

Business Processes to Improve:

Flight Operations

- To reduce flight delays to optimize passenger experience with on-time performance.

Revenue Growth Strategy:

- Analyze existing revenue streams and identify areas for improvement.
- Develop personalized pricing models to incentivize travel during non-peak periods.
- Implement dynamic pricing strategies based on demand forecasting.
- Explore partnerships with hotels, travel agencies, or other complementary services to create bundled offerings.

Marketing Campaign Execution:

- Perform a comprehensive analysis of customer behavior during non-peak seasons.
- Segment customers based on demographics, preferences, and behaviors. to tailor marketing messages and promotions.
- Utilize data analytics to identify the most effective channels for reaching target audiences.
- Implement digital marketing campaigns, social media promotions, and loyalty programs to encourage customer engagement.

B) Acquired data source:

1. Flight Delay Data for U.S. Airports by Carrier (Aug 2013 - Aug 2023) (* Primary Data source)
https://www.kaggle.com/datasets/sriharshaeedala/airline-delay
(CSV format)
https://drive.google.com/file/d/1WzhMFCcYuEgWu5z7vIF71BU2i8srUWNf/view?usp=sharing

Airline_Delay_Cause.csv (28.73 MB)

Detail Compact Column 10 of 21 columns

# year	# month	A carrier	A carrier_na...	A airport	A airport_na...	# arr_flights	# arr_del15	# carrier_
2023	8	9E	Endeavor Air Inc.	ABE	Allentown/Bethlehem/Easton, PA: Lehigh Valley International	89.00	13.00	2.25
2023	8	9E	Endeavor Air Inc.	ABY	Albany, GA: Southwest Georgia Regional	62.00	10.00	1.97
2023	8	9E	Endeavor Air Inc.	AEX	Alexandria, LA: Alexandria International	62.00	10.00	2.73
2023	8	9E	Endeavor Air Inc.	AGS	Augusta, GA: Augusta Regional at Bush Field	66.00	12.00	3.69
2023	8	9E	Endeavor Air Inc.	ALB	Albany, NY: Albany International	92.00	22.00	7.76
2023	8	9E	Endeavor Air Inc.	ATL	Atlanta, GA: Hartsfield-Jackson Atlanta International	1636.00	256.00	55.98
2023	8	9E	Endeavor Air Inc.	AUS	Austin, TX: Austin - Bergstrom International	75.00	12.00	5.62
2023	8	9E	Endeavor Air Inc.	AVL	Asheville, NC: Asheville Regional	59.00	7.00	3.32
2023	8	9E	Endeavor Air Inc.	AZO	Kalamazoo, MI: Kalamazoo/Battle Creek International	62.00	13.00	6.53

2. US Weather Events (2016 - 2022)
https://www.kaggle.com/datasets/sobhanmoosavi/us-weather-events/data
(CSV format)
https://drive.google.com/file/d/1duWT-S_VY664q_wJKxYU4dGkHkKyolF6/view?usp=sharing

WeatherEvents_Jan2016-Dec2022.csv (1.09 GB)

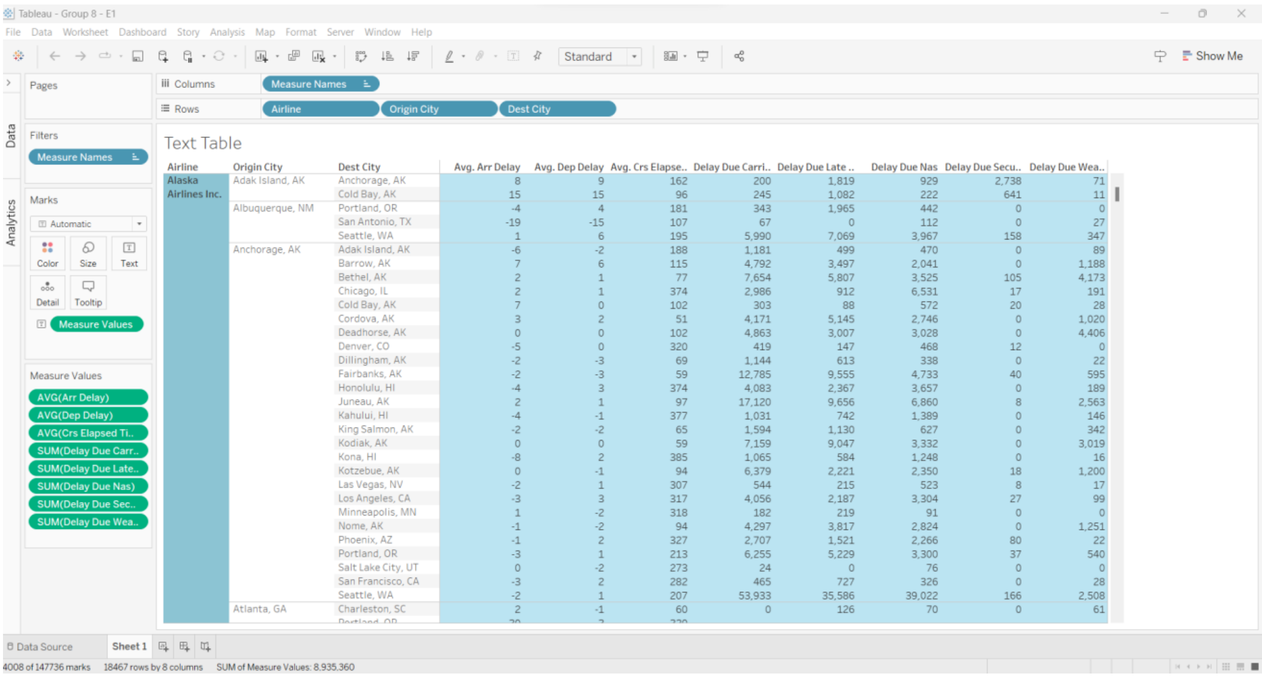
Detail Compact Column 10 of 14 columns

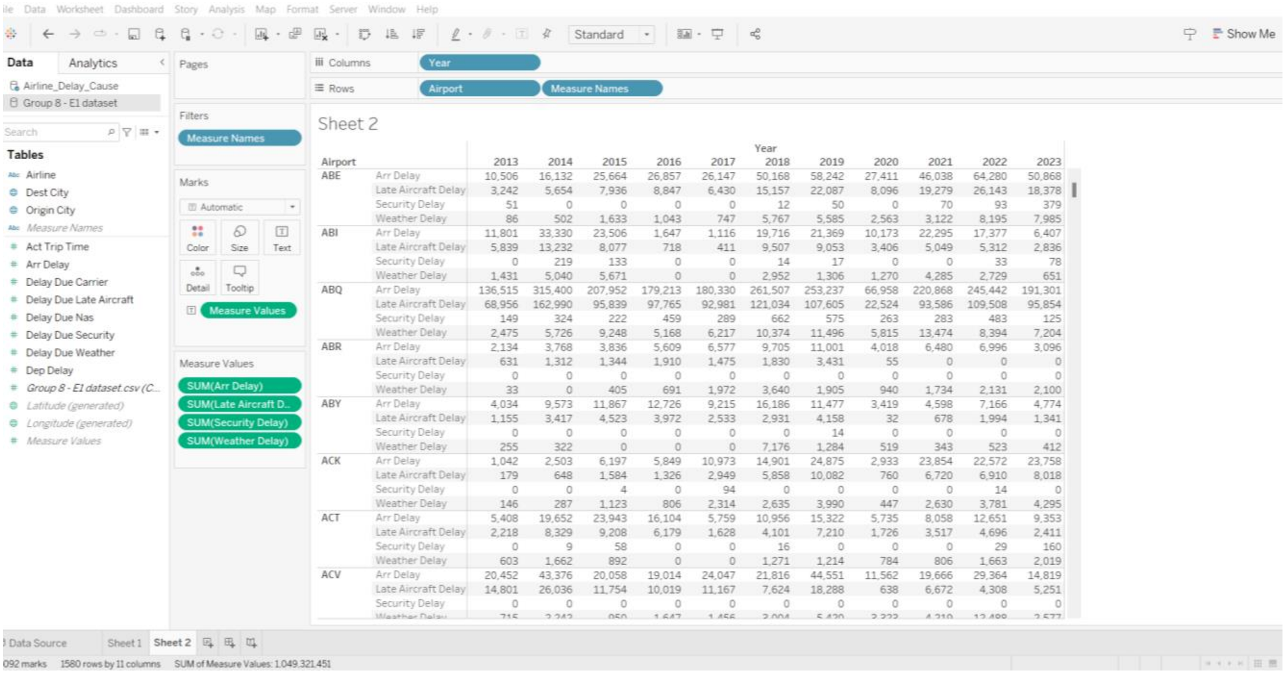
A EventId	A Type	A Severity	StartTime(UT...	EndTime(UTC)	# Precipitation(i...	A TimeZone	A AirportCode	# LocationLat	# LocationLng
W-1	Snow	Light	2016-01-06 23:14:00	2016-01-07 00:34:00	0.0	US/Mountain	K04V	38.0972	-106.1689
W-2	Snow	Light	2016-01-07 04:14:00	2016-01-07 04:54:00	0.0	US/Mountain	K04V	38.0972	-106.1689
W-3	Snow	Light	2016-01-07 05:54:00	2016-01-07 15:34:00	0.03	US/Mountain	K04V	38.0972	-106.1689
W-4	Snow	Light	2016-01-08 05:34:00	2016-01-08 05:54:00	0.0	US/Mountain	K04V	38.0972	-106.1689
W-5	Snow	Light	2016-01-08 13:54:00	2016-01-08 15:54:00	0.0	US/Mountain	K04V	38.0972	-106.1689
W-6	Snow	Light	2016-01-08 16:14:00	2016-01-08 17:34:00	0.0	US/Mountain	K04V	38.0972	-106.1689
W-7	Fog	Severe	2016-01-09 12:54:00	2016-01-09 15:34:00	0.0	US/Mountain	K04V	38.0972	-106.1689
W-8	Snow	Light	2016-01-09 15:34:00	2016-01-09 16:14:00	0.0	US/Mountain	K04V	38.0972	-106.1689
W-9	Fog	Severe	2016-01-09 16:14:00	2016-01-09 16:34:00	0.0	US/Mountain	K04V	38.0972	-106.1689
W-10	Snow	Light	2016-01-09 16:34:00	2016-01-09 16:54:00	0.0	US/Mountain	K04V	38.0972	-106.1689
W-11	Cold	Severe	2016-01-09 16:54:00	2016-01-09 20:34:00	0.0	US/Mountain	K04V	38.0972	-106.1689
W-12	Fog	Severe	2016-01-10 02:54:00	2016-01-10 04:14:00	0.0	US/Mountain	K04V	38.0972	-106.1689
W-13	Fog	Severe	2016-01-10 09:34:00	2016-01-10 10:14:00	0.0	US/Mountain	K04V	38.0972	-106.1689

3. Average Domestic Airline Itinerary Fares (1993 -2023)
https://www.transtats.bts.gov/averagefare/
(CSV format)
https://drive.google.com/file/d/1M6Fcedcn5Nn7lJZoVDHo8JIh8g0jwV2Q/view?usp=sharing

2022 Passenger Rank	Airport Code	City Name	State Name	Average Fare ($)	Inflation Adjusted Average Fare ($) (Base Quarter: Q2-2023)
1	LAX	Los Angeles	CA	424.87	424.87
2	ORD	Chicago-O'Hare	IL	390.64	390.64
3	DEN	Denver	CO	360.53	360.53
4	ATL	Atlanta	GA	397.90	397.90
5	EWR	Newark	NJ	411.70	411.70
6	BOS	Boston	MA	401.23	401.23
7	SEA	Seattle	WA	409.93	409.93
8	MCO	Orlando	FL	271.86	271.86
9	DFW	Dallas-DFW	TX	423.78	423.78
10	PHX	Phoenix	AZ	392.05	392.05
11	SFO	San Francisco	CA	466.42	466.42
12	JFK	New York-JFK	NY	421.94	421.94
13	LAS	Las Vegas	NV	273.47	273.47
14	LGA	New York-La Guardia	NY	332.56	332.56
15	MSP	Minneapolis	MN	421.57	421.57
16	PHL	Philadelphia	PA	419.69	419.69
17	IAH	Houston-Intercontinental	TX	420.45	420.45
18	SAN	San Diego	CA	390.26	390.26
19	DTW	Detroit	MI	429.18	429.18
20	FLL	Fort Lauderdale	FL	272.93	272.93
21	DCA	Washington-Reagan National	DC	399.08	399.08
22	TPA	Tampa	FL	335.18	335.18
23	MIA	Miami	FL	346.24	346.24
24	AUS	Austin	TX	380.51	380.51

C) Consolidated Tableau Packaged Workbook:
https://drive.google.com/file/d/11dPRVvrYAsGv8E02GRt2w3twoYO984_M/view?usp=sharing





Sheet 2

Airport		2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023
ABE	Arr Delay	10,506	16,132	25,664	26,857	26,147	50,168	58,242	27,411	46,038	64,280	50,868
	Late Aircraft Delay	3,242	5,654	7,936	8,847	6,430	15,157	22,087	8,096	19,279	26,143	18,378
	Security Delay	51	0	0	0	0	12	50	0	70	93	379
	Weather Delay	86	502	1,633	1,043	747	5,767	5,585	2,563	3,122	8,195	7,985
ABI	Arr Delay	11,801	33,330	23,506	1,647	1,116	19,716	21,369	10,173	22,295	17,377	6,407
	Late Aircraft Delay	5,839	13,232	8,077	718	411	9,507	9,053	3,406	5,049	5,312	2,836
	Security Delay	0	219	133	0	0	14	17	0	0	33	78
	Weather Delay	1,431	5,040	5,671	0	0	2,952	1,306	1,270	4,285	2,729	651
ABQ	Arr Delay	136,515	315,400	207,952	179,213	180,330	261,507	253,237	66,958	220,868	245,442	191,301
	Late Aircraft Delay	68,956	162,990	95,839	97,765	92,981	121,034	107,605	22,524	93,586	109,508	95,854
	Security Delay	149	324	222	459	289	662	575	263	283	483	125
	Weather Delay	2,475	5,726	9,248	5,168	6,217	10,374	11,496	5,815	13,474	8,394	7,204
ABR	Arr Delay	2,134	3,768	3,836	5,609	6,577	9,705	11,001	4,018	6,480	6,996	3,096
	Late Aircraft Delay	631	1,312	1,344	1,910	1,475	1,830	3,431	55	0	0	0
	Security Delay	0	0	0	0	0	0	0	0	0	0	0
	Weather Delay	33	0	405	691	1,972	3,640	1,905	940	1,734	2,131	2,100
ABY	Arr Delay	4,034	9,573	11,867	12,726	9,215	16,186	11,477	3,419	4,598	7,166	4,774
	Late Aircraft Delay	1,155	3,417	4,523	3,972	2,533	2,931	4,158	32	678	1,994	1,341
	Security Delay	0	0	0	0	0	0	14	0	0	0	0
	Weather Delay	255	322	0	0	0	7,176	1,284	519	343	523	412
ACK	Arr Delay	1,042	2,503	6,197	5,849	10,973	14,901	24,875	2,933	23,854	22,572	23,758
	Late Aircraft Delay	179	648	1,584	1,326	2,949	5,858	10,082	760	6,720	6,910	8,018
	Security Delay	0	0	4	0	94	0	0	0	0	14	0
	Weather Delay	146	287	1,123	806	2,314	2,635	3,990	447	2,630	3,781	4,295
ACT	Arr Delay	5,408	19,652	23,943	16,104	5,759	10,956	15,322	5,735	8,058	12,651	9,353
	Late Aircraft Delay	2,218	8,329	9,208	6,179	1,628	4,101	7,210	1,726	3,517	4,696	2,411
	Security Delay	0	9	58	0	0	16	0	0	0	29	160
	Weather Delay	603	1,662	892	0	0	1,271	1,214	784	806	1,663	2,019
ACV	Arr Delay	20,452	43,376	20,058	19,014	24,047	21,816	44,551	11,562	19,666	29,364	14,819
	Late Aircraft Delay	14,801	26,036	11,754	10,019	11,167	7,624	18,288	638	6,672	4,308	5,251
	Security Delay	0	0	0	0	0	0	0	0	0	0	0
	Weather Delay	715	[illegible]	[illegible]	1,647	1,456	[illegible]	[illegible]	[illegible]	[illegible]	[illegible]	2,577